

Express.js — Day 5 Notes

JSON Responses & HTTP Status Codes

1. What We Learned

Day 5 focused on sending structured JSON responses and choosing the correct HTTP status code for the result of a request.

2. res.json()

res.json() sends structured data as JSON. We can pass a JavaScript object to it; Express serializes that object into JSON for the HTTP response.

```
app.get('/users', (req, res) => {  
  res.json({  
    id: 25,  
    name: 'Bhavana',  
    role: 'Developer'  
  })  
})
```

The JavaScript object is not literally the same text as JSON. The browser receives JSON text such as:

```
{"id":25,"name":"Bhavana","role":"Developer"}
```

3. res.send() vs res.json()

res.send() is a general-purpose response method and can send text, HTML, buffers, and other values. **res.json()** is specifically intended for JSON API responses and clearly communicates that the response is JSON.

Note: **res.send()** can also receive an object and Express can serialize it. When you intentionally return API data as JSON, prefer **res.json()**.

4. Setting Status Codes

Use **res.status(code)** to set the HTTP status, then send the response.

```
res.status(200).json({  
  message: 'Success'  
})
```

Code	Meaning	Think / Use
200	OK	Request succeeded
201	Created	A new resource was created
400	Bad Request	Client sent an invalid request
401	Unauthorized	Authentication/login is required
403	Forbidden	User is not allowed
404	Not Found	Resource/route was not found
500	Internal Server Error	Server-side problem

5. 401 vs 403

401 Unauthorized → “Who are you?” Login/authentication is required.

403 Forbidden → “I know who you are, but you are not allowed.” The user lacks permission.

6. Practical Examples

```
res.status(200).json({ message: 'Success' })
res.status(201).json({ message: 'User created' })
res.status(400).json({ error: 'Name is required' })
res.status(401).json({ error: 'Please login' })
res.status(403).json({ error: 'Access denied' })
res.status(404).json({ error: 'User not found' })
res.status(500).json({ error: 'Something went wrong' })
```

7. Day 5 Assessment — 5/5 Correct

401 → authentication required

403 → authenticated but no permission

400 → invalid request

404 → resource not found

500 → server/internal failure

Day 5 DONE ■

Next: Day 6 — Request & Response in more depth.